

New York City Sustainability and Climate-Related Disasters Overview Memo

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New York City's sustainability and resilience landscape is characterized by an extensive history of climate shocks, complex and dense infrastructure needs, and a constantly changing network of detailed plans. These plans are designed to address both immediate issues and ongoing problems, as well as to support long-term structural planning to address vulnerabilities. Throughout the course, we learned about the importance of planning, resilience, and critical infrastructure systems. In our memo, we analyze how NYC's planning documents highlight how the city responds to societal threats related to climate, infrastructure, and inequality. New York City has faced numerous issues, including extreme rainfall and coastal exposure, as well as fossil fuel dependence and inequitable planning. The city's expansive infrastructure system that implements stormwater drainage, metro flood defenses, electrical grids, and waste management shapes how the city has vulnerabilities, but also the ability to adapt. We can better understand how NYC's disaster history has influenced policy changes and plans put in place to address sustainability and build resilience.

City Plans

New York City's sustainability and disaster planning pursues comprehensive visions of future resiliency and justice, along with specific measures to adapt to major coastal hazards and improve key stormwater and waste management infrastructures.

OneNYC 2050 is its comprehensive plan to achieve a stronger, fairer city with a diverse range of objectives, including providing a livable climate by generating net-zero electricity and strengthening communities and infrastructure in flood-prone areas (City of New York, Mayor

Bill De Blasio, 2019). The aim to advance social justice through climate adaptation measures and infrastructure improvement reflects a view discussed in the lecture of the city as a system of interacting social, environmental, and technological features that can mutually reinforce each other to achieve a sustainable balance. Meanwhile, NYC's Stormwater Resilience Plan is oriented toward addressing one key climate risk, flooding, by reducing vulnerability. It calls for raising awareness of vulnerability with mapping and updating flash flood response procedures to better protect vulnerable communities, such as by evaluating key storm shelter areas (NYC Mayor's Office of Resiliency, 2021). Through its focus on reducing vulnerability via public engagement and enhancement of emergency infrastructures, the plan reflects the need to address flooding and promote equity through social, environmental, and technological systems.

Furthermore, NYC's Hazard Mitigation Plan seeks to reduce vulnerability to not only flooding but all of its major hazards. Its goals are to protect public health and safety, property, and generate economic capacity for mitigation (NYCEM, 2025). They are to be met by improving communication with vulnerable communities, considering risk information in building codes and siting, and developing partnerships with private businesses to invest in hazard mitigation (NYCEM, 2025). By aiming to alter the siting of buildings and invest in preventative measures, the plan reduces risk not only by adapting to reduce vulnerability but also by mitigating hazards and exposure. Additionally, NYC's Draft Solid Waste Management Plan for 2026 supports reducing waste generation and pollution from special waste (DSNY, 2025). It features a program to promote a circular economy for the reuse of materials and a program to expand community events for special waste collection (DSNY, 2025). Like the Hazard Mitigation Plan, the waste management plan emphasizes empowering the general public and the

private sector to play a role in mitigating climate hazards through waste and emissions reductions, leveraging social components to improve the urban environment.

Across all four plans, there is a focus on adapting to and mitigating flooding by strengthening critical infrastructure or empowering the public to reduce greenhouse gas emissions that contribute to climate change. That focus is understood as a response to increasing flood risk and the challenges it poses to NYC's aging networks of critical infrastructure.

Critical Infrastructure

NYC is known for its public transportation that includes subway systems, buses, grid-like streets, large bridges, tunnels, ferries, and airports. The Metropolitan Transportation Authority (MTA) oversees operations in subways and railroads. The Department of Transportation (DOT) operates with buses, 6300 miles of roads, and over 800 bridges, where lastly the Port Authority operates with airports and ship ports such as JFK and LaGuardia (NYC.gov DOT). Most of the existing infrastructure in the city is over a hundred years old, with the first subway line opening in 1904 and numerous tunnels and bridges being built in the early 20th century. The maintenance needed for the services includes track and tunnel repairs, safety inspections, and road repairs. The repairs come with a great cost exceeding \$19 billion, supported by the annual budget that gets revenue from fares, tolls, bonds, and federal grants (NYU Wagner, 2013). The existing aging infrastructure is very vulnerable to climate risks, and as seen during Hurricane Sandy, there was flooding in tunnels and road closures that hindered operations. The levels of damage exposed vulnerabilities in their disaster response system and highlighted the need for improved public transportation services that are resilient. Its criticality for residents stems from the need for mobility, the economic activity it brings, and dependence on reliable energy, water, and communications.

Being one of the largest cities in the world comes with an extensive need for waste removal and management. NYC's waste infrastructure operates through curbside collection, transfer stations, landfills, recycling plants, and destruction plants. The primary municipality that controls these services is the Department of Sanitation (DSNY), which is responsible for moving waste to transfer stations before being sent out to landfills or recycling plants. Regular waste and trash are taken by licensed private companies or municipalities (NYC.gov DSNY). The larger trash facilities are generally privately owned, while the majority of them are publicly funded; each facility is regulated by the NYC Department of Environmental Conservation (dec.ny.gov). There are significant maintenance requirements such as street cleaning, vehicle maintenance, and facility inspections. The cost of maintenance is very high, with over \$1.6 billion coming from tax revenue and \$730 million from private businesses (cbcny.org). The current state of the infrastructure has around 16-25 years left of remaining capacity where after which they will need expansion. There have been issues with regulation where buildings with more than 9 units must dispose of waste bags outside, causing problems with rat infestations (NY Times). Waste management infrastructure is essential for public health and the general ecosystem of the city's sustainability practices.

Housing in NYC is a blend of private rental and owner-occupied units, where the public housing is managed by the New York City Housing Authority (NYCHA). The affordable housing units are supported by the Department of Housing Preservation & Development (HPD) (nyc.gov HPD). The housing is some of the oldest in the USA, with more than 80% of the housing units being over 50 years old (CHPCNY.org). It's characterized by high maintenance costs and repairs, including roofs, windows, elevators, and plumbing. NYCHA reports that there are tens of billions in unmet capital needs, and they cite around 40-70 billion needed for

multi-year growth (NYCHA, 2023). Funding comes from rent, city budget allocations, federal programs, and private-public partnerships. The housing infrastructure relies on services such as gas, electricity, and water, which are operated by other agencies in the city, highlighting its interdependence on the other critical infrastructure systems.

NYC's infrastructure is aging and interconnected, which requires building resilience against climate risks. The infrastructure is undergoing modernization and resilience changes; however, it still faces major challenges in terms of budgeting, maintenance, adaptation, and equitable solutions to all communities.

Disaster Experience

New York City's disaster history shows how its coastal location, exposure to extreme weather, and aging infrastructure combine to cause widespread and unequal impacts. In recent decades, the city has faced repeated shocks, such as the 1977 blackout, major winter storms in 1993, 1996, and 2016, and heavy rainfall that overwhelmed drainage systems. However, Hurricanes Sandy in 2012 and Ida in 2021 had the greatest impact. These storms revealed how much the city relies on interconnected transportation, energy, stormwater, and housing systems, and how a failure in one can affect the others. Hurricane Sandy had the biggest impact of any recent disaster and changed how the city thinks about resilience. The storm caused tens of billions of dollars in damage, flooded subway and vehicle tunnels, knocked out power stations, and left large parts of the city without electricity. The MTA, Port Authority, and DOT had to shut down for long periods because of flooded tunnels, damaged tracks, and blocked roads, showing the weakness of the infrastructure built in the early 1900s. Public housing along the waterfront, much of it managed by NYCHA, went days or weeks without heat, elevators, or electricity. This showed how climate hazards make existing housing problems worse. Ida highlighted a different

but more common threat, extreme rainfall. Instead of a storm surge, the city faced heavy downpours that overwhelmed drainage systems, flooded basement apartments, and caused deaths in areas not usually considered high-flood risk. This event showed that New York's stormwater systems, many of which are over 100 years old, can't keep up with the heavier rainfall caused by climate change. It also showed that people living in informal or unsafe housing, like illegal basement units, are at greater risk. Taken together, these disasters show some clear patterns: when one part of the city's infrastructure fails, others often do too. Recovery is uneven across neighborhoods, and climate hazards now come in many forms, both coastal and inland. These lessons have shaped New York's planning. The Stormwater Resiliency Plan addresses flooding from heavy rain like Ida, the Hazard Mitigation Plan prepares for many types of hazards, and OneNYC 2050 connects climate adaptation with social justice. The city's experience shows that building resilience means upgrading infrastructure and reducing the inequalities that leave some communities more at risk.

Key Sustainability Challenges

New York City faces interconnected sustainability challenges driven both by accelerating climate hazards and long-standing infrastructural and government shortcomings. Increasingly intense rainfall events and rising sea levels put critical infrastructure in coastal communities at risk.

According to the NYC stormwater resilience plan, the heaviest 1% of rainfall events in the northeast have already increased by 70% since the mid-20th century, and projections suggest the city could experience up to 25% more annual rainfall by the end of the century. (NYC Stormwater Resiliency Plan, p.4) Because much of the city's stormwater system was designed using rainfall data from 1903-1951, it is not equipped to handle the modern-day extremes. (Stormwater Resiliency Plan, p.5). Along with rainfall, extreme heat is also a problem, as New

York's density of buildings, pavement, and lack of green space traps and intensifies heat, making the city hotter. The extreme heat disproportionately affects minority neighborhoods and low-income neighborhoods that tend to have fewer trees, less shade, and older buildings that lack proper air conditioning. Housing conditions worsen climate vulnerability as well: nearly 70% of NYCHA buildings were constructed before 1969, and over 42,000 public housing units need substantial repairs, and more than 500 NYCHA buildings are located in projected flood plains, exposing already vulnerable residents to multiple climate hazards at once. (EDF Report p.21)

The city's green infrastructure is also lacking. Although NYC has expanded bioswales, rain gardens, and other green stormwater systems, the Stormwater Resiliency Plan emphasizes the need for more targeted, neighborhood-specific expansion to manage the increasing volume and intensity of rainfall. These challenges are further heightened by governance and funding barriers. Both the Stormwater Resiliency Plan and the EDF report note that retrofitting aging infrastructure and upgrading flood protections require significant investment, and fragmentation across agencies slows implementation.

Conclusion

New York City is a unique system that requires tailored sustainability and disaster planning. It is socially diverse, environmentally susceptible to extreme rain and heat, and technologically dependent on infrastructure that major hurricanes have exposed as insufficient and inequitable. Yet, NYC's sustainability plans also look to infrastructure to overcome key challenges and improve equity. To forge a more resilient and just future, NYC must strengthen its hard and soft infrastructures: adapting stormwater infrastructure and flood protection, communicating with vulnerable communities, and mitigating hazards through a reduced waste stream.

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